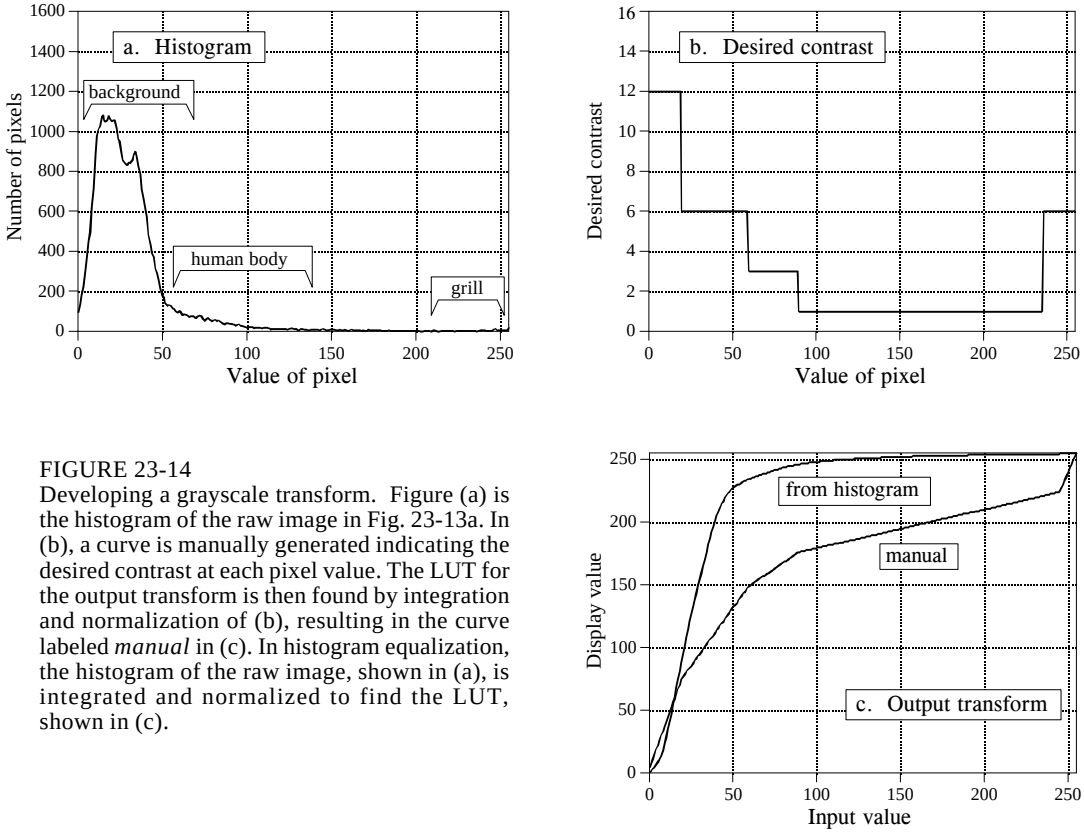

FIGURE 23-14
Developing a grayscale transform. Figure (a) is the histogram of the raw image in Fig. 23-13a. In (b), a curve is manually generated indicating the desired contrast at each pixel value. The LUT for the output transform is then found by integration and normalization of (b), resulting in the curve labeled *manual* in (c). In histogram equalization, the histogram of the raw image, shown in (a), is integrated and normalized to find the LUT, shown in (c).

right side. Why take the *integral* to find the required curve? Think of it this way: The contrast at a particular pixel value is equal to the slope of the output transform. That is, we want (b) to be the derivative (slope) of (c). This means that (c) must be the integral of (b).

Passing the image in Fig. 23-13a through this manually determined grayscale transform produces the image in (b). The background has been made *lighter*, the grill has been made *darker*, and both have better contrast. These improvements are at the expense of the body's contrast, producing a less detailed image of the intruder (although it can't get much worse than in the original image).

Grayscale transforms can significantly improve the viewability of an image. The problem is, they can require a great deal of trial and error. **Histogram equalization** is a way to automate the procedure. Notice that the histogram in (a) and the contrast weighting curve in (b) have the same general shape. Histogram equalization blindly uses the histogram as the contrast weighing curve, eliminating the need for human judgement. That is, the output transform is found by integration and normalization of the *histogram*, rather than a manually generated curve. This results in the greatest contrast being given to those values that have the greatest number of pixels.